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RESUME BUILDER, ANALYZER AND JOB RECOMMENDER PLATFORM

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ABSTRACT

In today's competitive job market, having a well-structured, optimized resume is essential for job seekers to stand out and secure desired positions. This Platform aims to streamline and enhance the resume building process while providing job seekers with valuable insights and recommendations. The platform offers three key features: the Resume Builder, which uses the Gemini API to generate content with minimal user input and AI-powered suggestions; the Resume Analyzer, which applies NLP and ML techniques to assess the resume against job descriptions and provide an ATS (Applicant Tracking System) score for optimization and the Job Recommender, which allows users to upload their resumes, extract information, and retrieve relevant job postings from job board APIs. By integrating AI, NLP, and ML technologies, the system helps users create optimized resumes, improve their interview chances, and find suitable job opportunities efficiently.

Keywords: Generative AI, Applicant Tracking System, Resume Builder, Resume Analyzer, Job Recommender.

I. INTRODUCTION

In today's highly competitive job market, job seekers face numerous challenges when it comes to creating impactful, industry-standard resumes and finding career opportunities that align with their unique skills and aspirations. The process of crafting a resume that is both ATS-compliant and professionally appealing often requires significant time, effort, and expertise—resources many job seekers may lack.[1] Existing resume-building tools and platforms often fall short of meeting these needs. They tend to focus narrowly on either design or content optimization without offering a holistic solution that integrates personalized features, in-depth resume analysis, and job recommendation functionalities.[2]

This fragmented approach leaves job seekers struggling to create resumes that stand out to recruiters and applicant tracking systems, often resulting in missed opportunities for interviews.[3] Additionally, the absence of tailored job recommendations further exacerbates the problem, as job seekers must sift through generic job postings that may not align with their qualifications.[4]

The goal of this project is to address these pain points by delivering a comprehensive and intelligent platform that simplifies and enhances every aspect of the job-seeking journey. By leveraging cutting-edge generative AI technologies, this solution aims to provide users with a seamless way to create high-quality, ATS-compliant resumes with minimal effort.[5] Furthermore, it will incorporate advanced analysis tools to evaluate resumes against industry standards, offering recommendations to optimize both content and structure.[6] Finally, the platform will integrate a personalized job recommendation system that aligns with the user's profile, bridging the gap between talent and opportunity. By uniting these features in a single, user-friendly platform, this project aspires to empower job seekers to achieve their career aspirations more efficiently and effectively.[8]

II. LITERATURE SURVEY

Recent advancements in NLP and ML have significantly improved job recommendation systems and resume screening technologies. Various ML-based systems now match applicants with job vacancies, showcasing the potential for automation in large-scale recruitment. Text mining techniques such as Term Frequency (TF) and TF-IDF are commonly used for job-resume matching, while personalized job recommendations benefit from feature selection methods and deep learning models like BERT and Named Entity Recognition (NER) for precise analysis [1][2]. Automated resume screening, powered by NLP and deep learning, has become a practical alternative to manual efforts. Approaches like keyword-based parsing, context-aware analysis, and integration with ATS improve recruitment accuracy and reduce bias [3][4]. Some systems employ Python-based resume parsers and PostgreSQL's trigram similarity for effective skill matching [5]. Advanced semantic search models within ATS frameworks improve search accuracy (up to 91.22%) but may struggle with scalability due to storage and indexing needs [6]. While some models segment resume content and use predefined taxonomies

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for job classification, they often lack adaptability and modern AI integration [7][8]. Machine learning methods, such as TF-IDF with cosine similarity and skill recommendation modules, enhance personalization by identifying skill gaps [9]. Deep learning frameworks using CNN and Random Forests recommend career paths with notable recall (84%) despite concerns over data bias and computational load [10]. Other systems apply Naïve Bayes for career suggestions, though reliance on LinkedIn data limits scope [11]. Content-based filtering and VSM approaches have also been used for candidate-job matching across skill sets, offering platform independence [13]. NER with SpaCy and Decision Trees has proven effective in extracting skills and matching candidates to roles [14]. Hybrid models combining content-based and collaborative filtering outperform traditional systems in terms of precision and recall [2]. Recent research highlights the value of AI in resume competitiveness evaluation, personality trait analysis, and personalized vacancy suggestions using TF-IDF and cosine similarity [15].

III. EXISTING SYSTEM

Traditional job recommendation and resume screening systems have relied heavily on rule-based filtering, keyword matching, and manual processes, which often result in inefficiencies and lack of personalization. The conventional approach focuses on matching predefined skills and job titles using basic string matching techniques, such as TF (Term Frequency) and TF-IDF (Term Frequency-Inverse Document Frequency), to represent resumes and job descriptions as document vectors. These techniques are widely used to compute similarity scores for job-resume matching but fail to capture the semantic context of the content. Many current platforms use basic Applicant Tracking Systems (ATS), which apply keyword-based filters to screen resumes. However, these systems often overlook qualified candidates due to lack of context-aware understanding, biases in keyword matching, and fixed resume formats. Resume parsers, such as Pyresparser, extract predefined fields like name, skills, and experience but do not analyze resume quality or relevance to job descriptions comprehensively. Furthermore, existing recommendation systems are usually limited to content-based or collaborative filtering models that do not incorporate the latest deep learning or NLP advancements. Although some platforms utilize machine learning techniques such as Naïve Bayes, Decision Trees, or cosine similarity, their performance is limited by shallow context understanding and the absence of dynamic job market integration.

IV. PROPOSED SYSTEM

The proposed system is an AI-powered platform designed to streamline the process of resume building, analysis, and job recommendations. It aims to overcome the limitations of existing systems by integrating Generative AI and intelligent automation.

The platform features an intelligent resume builder that leverages the **Llama3 large language model** to generate a professional and personalized resume. The user needs to input minimal details such as education, experience, and skills. The system uses generative capabilities to auto-generate well-structured content tailored to the user's domain and career goals, thereby saving time and ensuring quality.

The Resume Analyzer module evaluates the user's uploaded resume against the desired job description provided by the user. It uses the **Llama3 model** to semantically understand the content, compare it with industry expectations, and offer constructive feedback. This includes suggestions to improve formatting, keyword density, and content relevance.

To ensure the resume passes Applicant Tracking Systems (ATS) used by most recruiters, the system calculates an **ATS compatibility score**. The analyzer matches the resume's content against the job description and highlights missing skills or irrelevant sections that might reduce the selection probability, improving the candidate's chances of passing initial screenings.

The Job Recommender module connects with the **Jooble API** to fetch real-time job listings. Based on the resume content and user's preferences, the system uses AI matching logic to recommend jobs that align with the candidate's profile. This eliminates the manual effort of searching job boards and improves accuracy through LLM-based semantic matching.

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V. METHODOLOGY

Problem Identification

Identify the common challenges job seekers face in creating ATS-compliant resumes and finding jobs that align with their skills and aspirations. Highlight the inefficiencies of existing tools, which lack integration of resume building, evaluation, and job recommendation features. Define the goal of developing a unified platform to address these issues by using generative AI for resume creation and advanced analysis for job matching.

Data Collection and Preprocessing

Collect the personal information from users such as name, gender, education, projects etc., resume and job descriptions Apply preprocessing techniques such as tokenization, lemmatization, and stop-word removal to clean the data. Extract meaningful features like skills, experience, education and keywords using text mining techniques to structure the data for analysis and training machine learning models.

Resume Building

The Resume Builder collects resume-related information from the user, such as name, email, education, projects, skills, and work experience through an interactive form. Once the data is submitted, the system analyzes the provided information and formats it into a professionally structured resume. The generated resume follows industry standards to ensure readability and compatibility with Applicant Tracking Systems (ATS). The final resume is then made available for download in PDF format.

Resume Analyzer

The Resume Analyzer helps users assess the quality and relevance of their resumes against a specific Job Description (JD). The user uploads their resume and provides a JD as input. This data is then passed to the Generative AI model **LLama 3**, which evaluates the resume based on key factors such as **skill** relevance, missing keywords, and ATS compatibility. The system generates feedback that includes an **ATS score,** suggested improvements, and recommendations to enhance the resume for better job suitability.

Job Recommender

The Job Recommender simplifies the job search process by suggesting relevant job opportunities based on the candidate's resume. The user uploads their resume in PDF format, and the system extracts skills and key information using Generative AI. The extracted skills are then matched against available job listings fetched from a Jobs API. The platform compares the user's skill set with job requirements and retrieves the most relevant job postings, which are displayed to the user in an organized format.

Generative AI for Resume Building

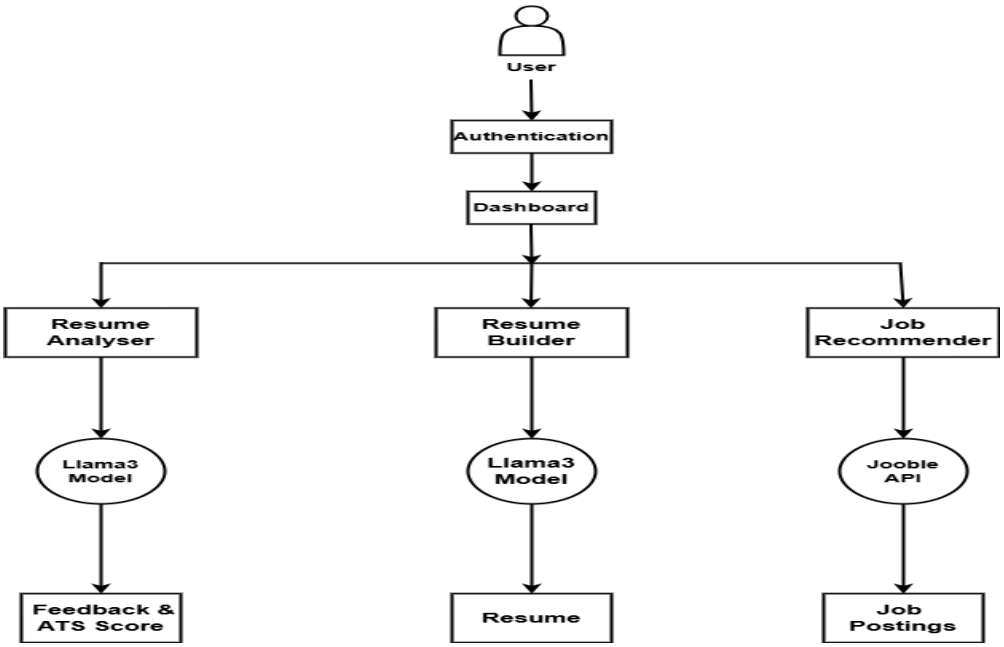


Figure 1: Proposed System Architecture.

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Develop a recommendation engine that matches user profiles with relevant job postings using techniques like cosine similarity, collaborative filtering, or machine learning models. Use the processed resume data to align user qualifications with job descriptions, prioritizing roles that match their skills, experience, and preferences. Provide tailored recommendations to bridge the gap between job seekers and suitable opportunities.

VI. CONCLUSION

The review paper presents a comprehensive AI-powered platform integrating resume building, analysis and Job recommendation. The proposed system presents a comprehensive solution to bridge the gap between job seekers and recruiters through AI-driven resume creation, intelligent resume analysis, ATS score evaluation, and job recommendations using real-time job data. By integrating advanced language models like LLaMA3 and APIs such as Jooble, the platform simplifies and automates the entire job application process. It not only enhances the quality of resumes but also improves the chances of selection by aligning them with job requirements. With potential future extensions like personalized interview preparation, AI-powered cover letters, and recruiter dashboards, this system can evolve into a complete career guidance platform, addressing the evolving needs of modern recruitment.

VII. FUTURE WORK

While the proposed system offers an intelligent and integrated approach to resume building, analysis, and job recommendation, there are several potential directions for future enhancement:

1. **Multi-Language Support:** Incorporating natural language processing models that support multiple regional and international languages would allow non-English-speaking users to benefit from resume generation and analysis in their preferred language.
2. **Integration with Multiple Job Boards:** Currently, the system uses the Jooble API. Future enhancements may include integration with other platforms like Indeed, LinkedIn Jobs, and Naukri.com to offer a more diverse and broader job pool.
3. **Smart Career Path Guidance:** Using predictive analytics and ML algorithms, the platform can recommend not only jobs but also suggest career paths, skill development courses, and certifications based on the user's interests, current resume strength, and market demand.
4. **Real-Time Chatbot Assistant:** Integrating a real-time AI chatbot using advanced LLMs like GPT-4 or Gemini Pro can assist users throughout the process—answering questions, suggesting better phrasing, or even offering interview preparation tips.
5. **Personalized Interview Question Generator:** Based on the user's resume and the job description, the system can generate a list of personalized interview questions using LLMs to help users prepare more effectively.
6. **Real-Time Market Trend Analysis:** Use job market data to show trending skills in different domains (e.g., Data Science, Web Development, Marketing) and suggest users update their resume accordingly.

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